

The Trap of AI Literacy: The Paradoxical Relationships Between College Students' Use of LLMs, AI Literacy, and Fact-checking Behavior

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Abstract

This study examines factors influencing users' critical engagement with large language models (LLMs), focusing on fact-checking behavior. While LLMs transform how individuals acquire knowledge, their rapid adoption raises concerns about uncritical acceptance due to limitations like hallucinations. A survey of college students and young professionals revealed nuanced effects of LLM literacy. Understanding LLM processes, such as input, processing, and output, encourages fact-checking. However, self-efficacy and knowledge of LLM features paradoxically reduce verification by fostering reliance on machine heuristics and elevating the perceived credibility of outputs. These findings highlight the complex role of AI literacy in promoting critical engagement and the importance of education to deepen users' technological understanding.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; **Heuristic evaluations**; **Natural language interfaces**.

Keywords

Artificial Intelligence, LLMs, AI literacy, Machine Heuristics, Credibility, Education

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1 Introduction

Large language models (LLMs) like ChatGPT, Gemini, and Claude have quickly gained global popularity for their intuitive chat-like interfaces and impressive performance. Their rapid adoption is transforming how people seek information and learn in daily work processes. The introduction of LLMs sparked a fierce debate on the technology's implication on higher education [25]. As an initial

response, universities quickly banned the use of ChatGPT or other similar LLMs in completing assignments, often without being able to implement concrete measures to prevent students from using the technology. Despite the higher education institutions' apparent hesitance to adopt LLMs in educational processes, a survey by Impact Research found that 82 percent of undergraduate students and 72 percent of K12 students surveyed had used AI tools for school work, such as writing assignments.

While popular LLMs such as ChatGPT or Claude exhibit an impressive ability to quickly generate relevant outputs in response to user prompts, they are not without limitations. LLMs often produce false and unsubstantiated information, referred to as "hallucinations" [24]. Therefore, users are encouraged to fact-check and cross-validate the outputs generated by LLMs [31]. However, due to the fluency and plausibility of the outputs, users may easily be convinced to trust the information provided by LLMs. This risk is heightened by the lack of proper knowledge and understanding of the technological processes underlying the technology – including learning and training, output generation, and their limitations and risks – which can lead users to overlook potential pitfalls. As LLMs are expected to be here to stay, increasing their influence on the way people seek new information and learn as shown by the statistics above, it is critical to enhance users' understanding of the technology as well as teach them ways to critically engage with LLMs.

Given this backdrop, this study identified factors related to LLM users' critical engagement with LLMs, specifically fact-checking of LLM-generated outputs. We found contradictory influences among different dimensions of LLM literacy. While an understanding of the technological principles of LLMs contributed to fact-checking behavior, self-evaluated confidence (i.e., self-efficacy) and knowledge of LLM features and functions undermined it by relying on heuristic shortcuts to overly trust LLM outputs. By examining the role of different dimensions of LLM literacy and cognitive processes, we illuminated the facilitators and barriers to the wise adoption of the technology, contributing to the broader discussion on AI literacy and responsible use of AI tools among young adults.

2 Related Works

2.1 LLMs and Fact-checking

A significant challenge in LLMs is their potential to generate unintentionally false but plausible outputs, widely known as hallucination [38]. Common examples of hallucination include the generation of unfounded references or unsubstantiated information [28].

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This issue is rooted in the way that LLMs are engineered to generate textual outcomes. LLMs are trained on enormous amounts of data to learn and understand the linguistic patterns observed from the data, which is called the pattern-generation technique (Ray, 2023). In response to user prompts, LLMs generate text outputs that are plausible within the given context and closely correspond to the observed pattern through complex algorithmic processes. However, without real-time monitoring and constant fine-tuning of the models, LLMs are prone to generating erroneous and factually inaccurate information as demonstrated in many existing studies [13, 33, 48]. For instance, according to a study that evaluated LLM factuality, the human-annotated error rate ranged from 9.4 to a whopping 68 percent depending on the type of dataset [49].

Existing research urges solutions to this vulnerability of LLMs as users increasingly rely on these tools for information search, retrieval, and learning [1]. While existing research is primarily geared toward hallucination mitigation techniques, scant attention has been paid to examining factors related to users' proactive engagement in fact-checking and output verification. For example, studies explore methods such as various prompt engineering tactics and model development [13, 38, 48]. Other studies test automatic fact-checking by developing a chatbot to discern misinformation [4, 10, 26].

While technological advancement is an important facet of addressing the issue of hallucination, we argue that another critical pillar should be enhancing users' awareness of it and encouraging them to critically engage with LLMs through fact-checking and verifying the information they produce. This study explores this topic by empirically identifying the factors associated with users' (non)engagement in fact-checking LLMs outputs to illuminate ways to mitigate the hallucination problem.

2.2 Predictors of Fact-checking Behavior

2.2.1 AI literacy. AI systems such as LLMs are often described as black boxes due to their highly complex back-end technological process. This complexity creates a significant gap in lay users' knowledge of how LLMs operate, making it increasingly challenging to have a clear and reasonable mental model [52]. Nonetheless, with the growing power of AI and its widespread application, enhancing users' literacy in this emerging technology is key to wise and critical adoption by users. Previous studies proposed that AI literacy is an essential competency encompassing both knowledge and skills for everyone in the era of intelligent technology, beyond those who are in closely related fields such as software developers, as AI is integrated into a wide variety of computing systems. These studies have defined AI literacy as one's ability to understand the basics of AI-driven computer systems and use them skillfully and ethically in everyday life settings [3, 14, 21].

In 2022, a UNESCO report on AI curricula for K-12 education highlighted that AI literacy equips students with the skills to comprehend the processes through which AI systems gather, process, and analyze data, as well as how AI algorithms identify patterns and relationships within the data [44]. Studies have also developed measures to assess one's AI literacy. For example, Ng and colleagues [21], conceptualized AI literacy in four domains: (1) knowing and understanding AI, (2) using and applying AI, (3) evaluating and

creating AI, and (4) ethics of AI, based on a pedagogical approach called ABCE framework [45]. Ng and colleagues emphasized that AI literacy should go beyond just knowing what AI is, and should encompass the ability to critically evaluate AI systems as well as the ability to work collaboratively with AI systems. In the following work, they developed an AI literacy scale based on the four dimensions [22]. In another noteworthy study [27], Pinski and Benlian pointed out that AI defies fundamental assumptions of previous conceptualization of user competencies in information systems literature, such as that humans are users, the way the technology works is comprehensible, and the operation of the technology is under full control by users or developers. Emphasizing these changing assumptions, they noted in their definition of AI literacy that it is important to incorporate users' understanding of the distinct facets of AI technology, as well as their ability to understand the steps of how AI handles input, processes information, and produces outputs.

Given these conceptualizations of AI literacy, those with greater literacy in AI, specifically LLMs, are not only more effective in leveraging the technology for a given task, but also more knowledgeable about the back-end processes of how the technology works, which allows them to understand the benefits as well as the limitations and risks of LLMs. Therefore, when using these tools on a daily basis, it is likely that those with higher literacy in LLMs are more likely to engage in fact-checking behavior because they recognize the technology's inherent limitations and the need to verify its outputs. Based on the discussion so far, we propose the following hypothesis regarding the relationship between individuals' literacy in LLMs and engagement in fact-checking during their routine use of these tools.

Hypothesis 1: Users with greater AI literacy will be more likely to engage in fact-checking behavior.

An important question to raise here is: what are the possible cognitive processes that underlie users with low LLM literacy then? We propose machine heuristics and perceived credibility as the key psychological variables that drive the uncritical acceptance of LLM outputs.

2.2.2 Machine Heuristics and Perceived Credibility of LLMs. Cognitive heuristics are a concept in cognitive psychology defined as mental shortcuts or rules of thumb often used to make a quick judgment or decision [42]. They serve as a powerful mechanism for decision-making and judgment through a distinctive way from deliberate and systematic thinking processes. For example, one might judge another person's credibility primarily based on cues invoked by their appearance or outfit (e.g., wearing a suit that gives an impression of professionalism and expertise), even though the judgment is not substantiated with other more concrete and relevant evidence.

Cognitive heuristics also apply to human-machine interaction. To identify cognitive heuristics at work during human-machine interaction, Yang and Sundar [36, 37, 50] proposed the concept called machine heuristics. Machine heuristics are activated when human users interact with computing systems such as LLMs. According to the HAIL-TIME model by Sundar [37], merely identifying AI as the source of information can trigger machine heuristics associated

with the technology. They consist of beliefs, such as machines are objective, accurate, efficient, and smarter than human counterparts in performing certain tasks.

These beliefs are derived from folk theories regarding “how machines work under the hood” [50]. From contemporary literature to Sci-Fi movies, and to modern philosophy [30, 41], machines are commonly depicted as possessing characters that are distinct from humans such as not having emotions, or being free from human limitations such as mental fatigue, biases or prejudice, or cognitive errors/malfunctions, which lead humans, particularly lay users, to expect them to be inherently more objective, accurate, and smarter in performing certain tasks than humans. Therefore, machine heuristics are positively associated with the perceived credibility of a technological system [36, 37, 50]. Perceived credibility is defined as users’ perception that a technology’s functionality and performance are trustworthy, expert, and reliable [19, 39].

Previous studies have demonstrated the positive relation between machine heuristics and the perceived credibility of various computing systems. For instance, news readers perceived news articles written or assisted by AI as more credible than those written by human authors only [47]. Similarly, a recent Pew Research Center study found that, while there is a mixed view on AI in healthcare, the American public in general believes AI would prevent errors made by human healthcare providers [43]. In a similar vein, these perceptions are also reflected in the concept called automation biases, which is the tendency to use “automation as a heuristic replacement for vigilant information seeking and processing” [20]. There has been considerable research on automation bias in domains such as aviation and clinical decision support systems [9, 35].

The common premise of these cognitive tendencies is that people rely on positive preconceptions about machines (i.e., machine heuristics) when interacting with computer systems, leading them to be inclined to trust systems without engaging in deep and thorough thinking processes. Applying this to the context of this study, we predict that individuals with lower AI literacy may be more prone to over-relying on surface-level impressions or the perceived authority of such systems given their limited understanding of how LLMs operate. Therefore, we predict that those with lower AI literacy are more likely to rely on machine heuristics when using LLM applications, compared to those with higher AI literacy, which then leads to a heightened perception of credibility toward LLMs. Therefore, we propose the following hypotheses.

Hypothesis 2: Users with lower AI literacy will exhibit a greater tendency toward machine heuristics.

Hypothesis 3: Positive machine heuristics will be positively associated with perceived credibility of LLMs.

2.3 Credibility and Fact-checking

The increased credibility attributed to LLMs due to machine heuristics may negatively impact users’ engagement in fact-checking behavior. The perceived credibility of technological systems plays a central role in shaping user attitudes and behavioral intentions toward technology, including its adoption and usage [39, 46]. A few previous studies under the context of the usage of ChatGPT

have found that trust or credibility toward ChatGPT is a crucial factor in adoption behavior. For instance, Shahzad et al. [32] found that higher levels of trust in ChatGPT positively influenced users’ attitudes and adoption intentions. Another study also found that trust in ChatGPT is a strong driver for usage, more than the perceived ease of use or usefulness of the technology [6]. While these studies show that credibility or trust attributed to LLM applications strongly predicts user adoption and positive attitudes, they do not reveal the credibility’s influence on how people may engage with LLM outputs. Based on the findings discussed above regarding machine heuristics and automation bias, we propose that heightened credibility through machine heuristics could create a false sense of security, reducing the perceived necessity for fact-checking. Therefore, we propose the following hypothesis:

Hypothesis 4: Perceived credibility of LLMs will be negatively associated with fact-checking.

Lastly, to illuminate the cognitive processes involved in the relationship between AI literacy and fact-checking behavior, the following serial mediation will be tested.

Hypothesis 5: In conclusion, AI literacy will positively impact fact-checking behavior through reduced reliance on machine heuristics and increased credibility.

3 Methodology

3.1 Study Design and Procedure

To examine the proposed hypotheses of this study, we conducted an online survey of LLM users who are either students currently attending college or higher education or young professionals in the U.S. and Singapore. Data was collected during the summer of 2024. U.S. participants were recruited through Prolific, while Singaporean participants were recruited via a research invitation posted on a public Telegram channel (i.e., SG Research Lobang). The survey procedure is as follows.

Upon giving consent to participate in the survey, participants were directed to the participation criteria question asking whether they have used any LLMs including ChatGPT, Gemini, Claud, or any other LLM services in the past few months. Because this study aimed to examine LLMs users’ fact-checking behavior, only those who answered yes to this question were able to continue with the survey. Those who met the criteria then answered questions about their engagement with fact-checking behavior while using LLMs, perceived credibility, self-reported AI literacy, and machine heuristics. Participants also provided information regarding the specific software they use, usage patterns, and their perception of LLMs, personal dispositions, as well as some open-ended questions regarding their experience with and perception of the LLMs. These variables were not the focus of this study and thus were not reported here. Lastly, participants provided their demographic information and ended the survey.

3.2 Study Participants

A total of 287 participants completed the survey in full and met the attention check criteria. 145 respondents were from the U.S. and 142 respondents were from Singapore. 124 participants were

male and 152 participants were female. The other 10 did not specify gender. The average age of respondents was 29.19 ($SD = 9.10$). 139 participants self-identified as Asian, 74 participants identified as White, 46 as Black, 13 as Hispanic, 8 as Others, 4 as American Indian or Alaska Native, and 1 as Native Hawaiian or Other Pacific Islander. In terms of the highest level of education, the majority held a Bachelor's degree ($N = 141$), followed by some college ($N = 54$), a Master's degree ($N = 35$), or High school graduates ($N = 31$). The remainder of the participants reported that they either had an associate degree ($N = 17$), a professional degree ($N = 3$), or a doctoral degree ($N = 1$), or received vocational training ($N = 3$) or some high school education with no diploma ($N = 1$).

3.3 Measures

3.3.1 AI Literacy. To measure participants' AI literacy, specifically regarding LLMs, we adopted two different AI literacy scales developed in previous studies [22, 27] and slightly modified them to meet the purpose of this study. The survey items included four sub-dimensions of AI literacy, namely, "self-efficacy," "cognitive learning," "AI technology knowledge," and "AI steps knowledge."

The first two measures were adopted from Ng and colleague's study and the last two measures were adopted from Pinski and colleagues' study. Self-efficacy (SE hereafter; $M = 5.11$, $SD = 1.28$, $\alpha = .92$) is defined as one's self-belief in their capability to perform a given task successfully, reflecting affective learning [2]. Specific items included: "I am confident I will perform well on LLM-related tasks." Cognitive learning (CL hereafter; $M = 5.07$, $SD = 1.33$, $\alpha = .82$) encompasses both lower-order and higher-order thinking skills related to LLMs. Lower-order skills refer to one's basic knowledge of LLM definitions and use cases (e.g., "I know what LLMs are and their definition"), while higher-order skills involve one's ability to evaluate, apply, and create LLM-driven solutions, such as appraising LLM-generated outputs or designing LLM-driven applications. Survey items included: "I know what LLMs are and their definition" and "I can apply LLM applications to solve problems." While SE captures one's confidence in their abilities, CL reflects the extent of one's AI-related knowledge and problem-solving skills. Thus, together, these two dimensions represent distinct but complementary aspects of an individual's perceived ability to engage with LLM applications.

While the first two dimensions focus on individuals' confidence in their ability to use LLMs and their knowledge of how to apply them, the other two measures focused primarily on measuring respondents' understanding of back-end technical processes underlying LLM operation. AI tech knowledge sub-dimension (TK hereafter; $M = 4.29$, $SD = 1.55$, $\alpha = 0.88$) included items such as "I have knowledge of the types of technology that LLM is built on," and "I have the knowledge of how LLMs and other AI technologies are distinct." AI steps knowledge sub-dimension (SK hereafter; $M = 4.15$, $SD = 1.65$, $\alpha = .95$) included items such as "I have the knowledge of the input data requirements for LLMs," "I have the knowledge of AI processing methods and models," and "I have knowledge of LLMs processing methods and models."

3.3.2 Machine Heuristics. For machine heuristics [50] three subdomains of machine heuristics, namely accuracy ($M = 4.89$, $SD = 1.49$, $\alpha = .90$), objectivity ($M = 4.52$, $SD = 1.69$, $\alpha = .96$), and expertise

($M = 5.44$, $SD = 1.45$, $\alpha = .90$), were used in the survey. These three measures were selected as they are closely related to the source credibility perception of LLMs. Respondents indicated their agreement with adjectives that describe the characteristics of LLMs, such as "LLMs are accurate," "LLMs are knowledgeable," "LLMs are unbiased," and "LLMs are smart." A total of 12 adjectives were used to measure the three subdomains of positive machine heuristics.

3.3.3 Credibility. To assess credibility, respondents were asked to indicate how credible they perceive the LLMs they use, using the prompt: "In general, how credible do you perceive the LLMs you use?" We employed four semantic differential items: "not credible - credible," "unreliable - reliable," "not expert - expert," and "not trustworthy - trustworthy." The four items showed good reliability ($M = 4.95$, $SD = 1.40$, $\alpha = .90$).

3.3.4 Fact-checking Behavior. To measure fact-checking behavior, we adopted and modified six items used in previous studies [5, 15, 23, 51] examining online information fact-checking behaviors ($M = 4.57$, $SD = 1.61$, $\alpha = .87$). Participants were asked to respond to the question "when using LLMs, how often do you engage in the following behaviors?" by indicating their engagement with fact-checking behaviors such as verifying the credentials or qualifications of authors or sources of information, consulting multiple sources to verify the information, using other search engines to further research on the topic, and seeking opinions or expertise of other individuals.

4 Results

4.1 Model Fit

The propose hypotheses were tested through path analysis using R lavaan packages. The path analysis model was assessed using maximum likelihood (ML) estimation with 1,000 bootstrap samples to obtain robust standard errors. The model demonstrated a good fit to the data. The chi-square test for the model fit was non-significant, $\chi^2(3)=3.710$, $p = 0.30$, meaning that the hypothesized model aligns well with the observed data. CFI (comparative fit indices) and TLI (Tucker-Lewis Index) were 0.999 and 0.993 respectively, showing excellent model fit [12, 16, 17, 40]. RMSEA (The Root Mean Square Error of Approximation) was 0.03 and SRMR (Standardized Root Mean Square Residual) was 0.01, suggesting a strong model fit. Together, these indices confirmed that the hypothesized path model provides a well-fitting representation of the data.

4.2 AI Literacy and Fact-checking

H1 predicted that individuals with higher AI literacy would be more likely to engage in fact-checking behavior. AI technology knowledge (TK) ($\beta = -0.09$, $p = 0.37$), self-efficacy (SE) ($\beta = 0.12$, $p = 0.13$), and cognitive learning dimensions (CL) ($\beta = 0.10$, $p = 0.35$) were not significantly related to the fact-checking behavior, while AI steps knowledge (SK) was significantly associated with the engagement in the fact-checking behavior ($\beta = -0.09$, $p = 0.37$). Therefore, H1 was partially supported.

4.3 Machine Heuristics, Credibility, and Fact-checking

H2 predicted that those with lower AI literacy will have a greater tendency to exhibit positive machine heuristics. Of the four sub-dimensions of AI literacy, self-efficacy and cognitive learning, which reflect one's confidence in their fluency with the utilization of LLM tools, were positively associated with the three dimensions of machine heuristics. Specifically, SE and CL both showed significant positive associations with the perception that LLMs are experts (SE: $\beta = 0.24$, $p = 0.01$; CL: $\beta = 0.28$, $p = 0.02$), objective (SE: $\beta = 0.22$, $p = 0.05$; CL: $\beta = 0.37$, $p < .01$), and accurate (SE: $\beta = 0.25$, $p = 0.01$; CL: $\beta = 0.37$, $p < .01$). In contrast, AI Technology Knowledge and AI Steps Knowledge did not show statistically significant associations with any of these machine heuristics (see Table 1), suggesting that individuals with greater insight into how the back-end of LLMs work may not necessarily have positive machine heuristics, while those with higher self-efficacy and proficiency in utilizing LLM features and functions tend to hold positive machine heuristics.

Table 1: The Four Dimensions of LLM literacy on Machine Heuristics

DV	IV	β (SE)	P-value
MH Expert	Tech Knowledge	-0.14 (0.10)	.17
	Steps Knowledge	0.05 (0.10)	.59
	Self Efficacy	0.24 (0.09)	.01
	Cognitive learning	0.28 (0.12)	.02
MH Objective	Tech Knowledge	-0.11 (0.13)	.41
	Steps Knowledge	0.03 (0.13)	.79
	Self Efficacy	0.22 (0.11)	.05
	Cognitive learning	0.37 (0.13)	<.01
MH Accurate	Tech Knowledge	-0.19 (0.12)	.11
	Steps Knowledge	0.06 (0.12)	.60
	Self Efficacy	0.25 (0.10)	.01
	Cognitive learning	0.37 (0.12)	<.01

With regard to the associations between positive machine heuristics and the perceived credibility of LLMs (H4), the results showed strong and consistent associations between the three types of machine heuristics and credibility. Expertise had a strong positive effect on Credibility ($\beta = 0.35$, $p < 0.001$), along with Objectivity ($\beta = 0.10$, $p < .01$) and Accuracy ($\beta = 0.29$, $p < 0.001$). These findings suggest that positive machine heuristics enhance credibility assessments of LLM tools as well as content generated by LLMs. Further, the perceived credibility of LLMs showed a strong negative association with fact-checking behavior ($\beta = -0.30$, $p < 0.001$), confirming H4 which predicted that as the perceived credibility of LLMs increases, the likelihood of engaging in fact-checking decreases.

Finally, H5 proposed that AI literacy would positively impact fact-checking behavior through reduced machine heuristics and credibility. To answer this hypothesis, parallel serial mediation paths were examined in the path analysis model. Contrary to H5, we found a negative influence of SE and CL dimensions of AI literacy

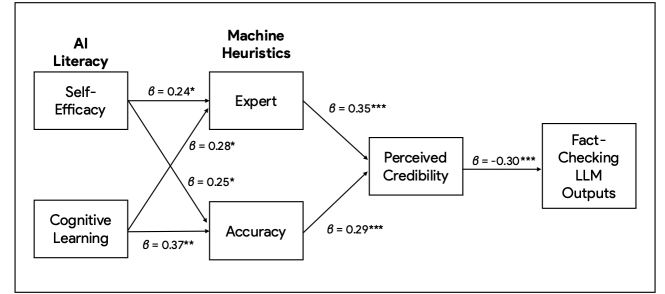


Figure 1: Serial mediation from SE and CL to factchecking. This figure only shows statistically significant paths, and is not a representation of the whole path analysis model

on fact-checking behavior. SE negatively influenced LLM users' fact-checking behavior via increased reliance on machine heuristics, specifically, expertise ($\beta = -0.03$, $p = 0.05$) and accuracy ($\beta = -0.02$, $p = 0.04$), both of which increased perceived credibility of LLMs, which then diminished fact-checking behavior. Similarly, CL reduced engagement in fact-checking via the same two dimensions of machine heuristics, expertise ($\beta = -0.03$, $p = 0.06$) and accuracy ($\beta = -0.03$, $p = 0.03$), which led to increased credibility and then decreased fact-checking behavior (see Figure 1).

5 Discussion

5.1 Summary of Key Findings

This study found a direct relation between individuals' AI steps knowledge, the knowledge of input, processing, and output processes of LLMs, and engagement in fact-checking behavior. However, ironically, two other dimensions of AI literacy—self-efficacy and cognitive learning—negatively influenced engagement in fact-checking behavior as they led to increased reliance on machine heuristics, specifically, the perceptions that machines are accurate and expert, which then elevated perceived credibility and reduced the need for verification of LLM outputs.

5.2 The Trap of AI Literacy

This study highlights the nuanced effects of different dimensions of AI literacy on users' engagement in fact-checking behavior when interacting with LLMs. Specifically, the positive direct impact of LLM steps knowledge on fact-checking behavior shows the importance of understanding the "under the hood" workings of LLMs. It encourages users to approach LLM outputs with a more critical lens, as they better recognize the inherent limitations, such as biases, inaccuracies, or errors of the technology. Conversely, the findings demonstrated that one's confidence in using LLMs effectively and knowledge of the functions and features of LLMs can diminish the likelihood of fact-checking. These users may be blind-sighted by the apparent sophistication and fluency of LLMs, resulting in "surface-level trust" that overlooks potential inaccuracies and reduces verification efforts.

In terms of conceptual understanding of AI literacy, the findings from this study call for a more nuanced approach, expanding on the existing discussion on digital literacy. Previous literature on digital

literacy primarily focuses on users' ability to utilize digital tools rather than their knowledge of the technological underpinnings of these tools [8, 11, 18]. However, our findings strongly suggest that the concept of AI literacy is multi-dimensional, and may not be equally beneficial in fostering wise and selective adoption of this technology. As we are entering the age of intelligent technology, it is crucial for users not only to be able to apply the technology to augment their productivity and performance but also, to discern how seeing intelligence is generated.

5.3 Implications on Higher Education and Critical Thinking

As LLMs become increasingly embedded in educational settings, many scholars have raised concerns regarding potential negative influences on learning processes and outcomes. For example, a few recent studies have found that using AI tools may reduce students' critical thinking skills and learning satisfaction [7, 29]. As the meaning of higher education lies not solely in gaining knowledge but also in developing critical thinking skills such as in-depth analysis and reflection, these findings are concerning.

Nonetheless, as technology's impact on higher education grows, it is crucial to integrate tools effectively while mitigating adoption risks. In this sense, our findings provide some meaningful contributions. Educational interventions fostering LLM literacy should prioritize "steps knowledge" which encourages a mindset of critical inquiry toward the technological tool. Merely teaching students functional knowledge and confidence can lead to unintended consequences, as observed by over-reliance on machine heuristics and unwarranted credibility.

This approach can enable students to understand the limitations, biases, and ethical implications of LLMs and to see beyond the impressive fluency and coherence of their outputs. At a more hands-on level, educators may consider training students to cross-validate LLM outputs with trustworthy sources, such as news articles, scholarly papers, and domain experts, and to identify any errors and biases embedded in them. Ultimately, the goal should be to cultivate an informed skepticism [34] that equips students with the ability to navigate the complex challenges of intelligent technologies responsibly.

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